

Critiquing Natural Language Processing in Healthcare:

Benefits, Limitations, and Future Implications

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Natural Language Processing (NLP) represents a transformative subset of artificial intelligence that focuses on enabling computers to understand, interpret, and generate human language. In healthcare settings, NLP technologies are increasingly being deployed to extract valuable clinical insights from electronic health records (EHRs), thereby addressing one of the most pressing challenges in modern medicine: making sense of the vast amounts of unstructured textual data generated during patient care. This report examines the application of NLP to medical records, evaluating its benefits and limitations while considering whether this technology ultimately serves the best interests of healthcare delivery and patient outcomes.

Understanding NLP and Its Application in Healthcare

NLP is a computational approach that enables machines to process and analyze large volumes of natural language data. In the healthcare context, NLP systems are designed to automatically scan through clinical notes, discharge summaries, pathology reports, and other narrative documentation within EHRs to identify and extract critical information such as diagnoses, medications, symptoms, and treatment outcomes (Kondylakis et al., 2023). Rather than requiring healthcare professionals to manually review hundreds of pages of patient records, NLP algorithms can rapidly locate relevant clinical data points, classify medical entities, and even predict patient risks based on patterns identified in the text.

Modern NLP systems in healthcare employ sophisticated techniques including deep learning models, particularly transformer architectures like BERT (Bidirectional Encoder Representations from Transformers), which can capture complex semantic relationships in clinical text. These systems have demonstrated robust performance in detecting conditions ranging from cognitive decline to cancer-related entities, with some models achieving sensitivity rates exceeding 90% (Kondylakis et al., 2023; npj Digital Medicine, 2025). The technology essentially serves as an

intelligent reading assistant that can process medical documentation at a scale and speed impossible for human reviewers alone.

Potential Benefits of NLP in Medical Records:

Enhanced Clinical Efficiency and Time Savings

One of the most significant benefits of NLP in healthcare is its capacity to dramatically reduce the time clinicians spend searching for information in patient records. Healthcare professionals often face the burden of reviewing extensive documentation to piece together a patient's clinical history, which can be both time-consuming and prone to oversight. NLP automates this labor-intensive process by rapidly extracting key clinical data points, allowing physicians and nurses to access critical information within seconds rather than hours. A comprehensive study analyzing over one million patients demonstrated that NLP successfully extracted and structured information from 157 million medical concepts across 9.5 million documents, highlighting its capability to process healthcare data at unprecedented scale (PLOS Digital Health, 2023). This efficiency gain translates directly into more time for patient care, as clinicians can redirect hours previously spent on documentation review toward direct patient interaction and clinical decision-making.

Improved Clinical Decision Support and Early Detection

Beyond efficiency, NLP systems offer substantial potential to enhance clinical decision-making by identifying patterns and risk factors that might otherwise remain hidden in unstructured text. These systems can automatically flag important clinical indicators, track disease progression over time, and even predict patient outcomes based on comprehensive analysis of clinical narratives. For instance, research has shown that NLP models can effectively detect early signs of cognitive impairment from routine clinical notes with median sensitivity of 0.88 and specificity of 0.96, enabling earlier intervention for conditions like dementia (npj Digital Medicine, 2025). Furthermore, NLP can extract psychosocial factors such as social isolation, housing insecurity, and financial stress from chart notes—information that, while documented, often goes unanalyzed

but can significantly impact patient health outcomes and hospitalization rates (PMC, 2024). By surfacing these insights systematically, NLP empowers clinicians to make more informed, holistic treatment decisions.

Limitations and Concerns Regarding NLP in Medical Records:

Privacy and Data Security Vulnerabilities

Perhaps the most pressing concern surrounding NLP deployment in healthcare involves patient privacy and the security of sensitive health information. NLP systems require access to vast quantities of patient data, including highly confidential clinical narratives that contain protected health information (PHI) such as patient names, addresses, medical conditions, and treatment histories. Recent research reveals troubling gaps in privacy protection: more than one-third of studies implementing NLP with patient data failed to report whether effective measures were in place to protect PHI, and nearly 10% neglected ethical review entirely (Journal of Medical Internet Research, 2025). The integration of EHRs with NLP technologies has significantly increased the risk of data breaches, with incidents ranging from ransomware attacks on health systems to unauthorized sharing of patient records with technology companies (Digital Health, 2025). Even well-intentioned NLP applications can inadvertently expose patient information through vocabulary dictionaries used in neural network training or through the unintended memorization of sensitive details within model parameters (Artificial Intelligence Review, 2022). These vulnerabilities are particularly concerning given that healthcare data breaches can expose not just current medical information but an individual's entire health history, creating lasting privacy implications.

Limited Generalizability and Data Quality Dependencies

A second critical limitation concerns the generalizability of NLP systems across different healthcare settings and their dependency on high-quality input data. NLP models are typically trained on specific datasets from healthcare organizations, and their performance can degrade

significantly when applied to data from different institutions, geographic regions, or patient populations. Studies consistently identify generalizability to other healthcare sectors, languages, and data sources as a major challenge, as NLP systems trained on one hospital's documentation style may perform poorly when deployed elsewhere (JAMIA Open, 2024). Moreover, the effectiveness of NLP is fundamentally constrained by the quality of the underlying clinical documentation. During periods of high clinical workload, such as during the COVID-19 pandemic, patient symptoms were often inadequately transcribed into records, resulting in skewed data that limited NLP's analytical capabilities (PMC, 2024). Incomplete documentation, inconsistent terminology, and variations in clinical note-taking practices across providers all introduce noise that can lead to inaccurate extractions and potentially dangerous clinical decisions based on flawed data analysis. Additionally, even high-performing NLP systems accumulate errors at scale: when processing millions of documents, even a low error rate can result in thousands of incorrect or missed clinical findings that might influence patient care.

Evaluating Whether NLP Benefits Healthcare

Despite the legitimate concerns outlined above, the application of NLP to medical records represents a fundamentally positive development for healthcare when implemented responsibly with appropriate safeguards. Technology addresses a genuine and growing problem: healthcare generates enormous volumes of valuable clinical information captured in free-text format, and without computational assistance, much of this information remains effectively inaccessible for systematic analysis or timely clinical use. The demonstrated ability of NLP systems to improve early disease detection, surface hidden risk factors, and save clinicians thousands of hours that can be redirected toward patient care constitutes tangible benefits that directly translate into better health outcomes.

However, realizing these benefits requires a measured approach that prioritizes patient privacy and data security. Healthcare organizations must implement robust privacy-preserving techniques such

as differential privacy, deidentification protocols, and federated learning approaches that enable NLP model training without centralizing sensitive patient data (npj Digital Medicine, 2025). Regulatory frameworks including HIPAA in the United States and GDPR in Europe provide essential guardrails, but enforcement must be rigorous, and organizations should adopt privacy protections that exceed minimum compliance requirements. Transparency is equally critical, patients deserve clear information about how their data is being used, and healthcare institutions should maintain oversight committees that include patient representatives in decisions about NLP deployment.

Regarding the generalizability challenge, the solution lies in collaborative development and validation. Networks like ENACT that enable multi-site NLP algorithm development and testing demonstrate how healthcare institutions can work together to create more robust, generalizable systems while maintaining local data control (PMC, 2025). Investment in standardized clinical documentation practices and ongoing model validation across diverse populations will gradually address performance disparities. Most importantly, NLP should be viewed as a decision-support tool that augments rather than replaces clinical judgment. Physicians must remain the final arbiters of clinical decisions, with NLP serving to surface relevant information more efficiently rather than autonomously directing care.

Conclusion:

Do the Benefits Outweigh the Risks?

After weighing the substantial benefits against the significant risks, the evidence suggests that the advantages of NLP in healthcare do outweigh the potential dangers, if implementation is accompanied by stringent privacy protections, rigorous validation, and appropriate human oversight. The alternative, continuing to allow vast amounts of clinically relevant information to remain trapped in unstructured text, inaccessible for systematic analysis or timely clinical use,

represents its own form of harm through missed opportunities for early intervention, unrecognized risk factors, and clinician burnout from information overload.

The key to successful NLP deployment lies in recognizing that technological capability must be matched with ethical responsibility. Privacy cannot be an afterthought, and generalizability cannot be assumed. Healthcare organizations must invest not only in NLP technology but also in the privacy infrastructure, validation frameworks, and oversight mechanisms necessary to deploy it safely. Patients must be informed partners in this process, with clear consent mechanisms and transparent communication about data use.

Looking forward, the continued evolution of privacy-preserving techniques such as federated learning and differential privacy offers promising pathways to harness NLP's benefits while minimizing risks. The healthcare community's growing recognition of these challenges, evidenced by increasing attention to privacy considerations in recent research, suggests a maturing field that is learning to balance innovation with responsibility. Ultimately, when implemented with appropriate safeguards and oversight, NLP in healthcare represents not a dangerous experiment but rather an essential evolution in how we extract value from clinical data to improve patient care. The question is not whether to use NLP in healthcare, but rather how to use it responsibly—and on that question, the path forward requires continued vigilance, investment in privacy protection, and unwavering commitment to keeping patient welfare at the center of all technological advancement.

References

- Conduah, A. K., Ofoe, S., & Siaw-Marfo, D. (2025). Data privacy in healthcare: Global challenges and solutions. *Digital Health*, 11, Article 20552076251343959. <https://doi.org/10.1177/20552076251343959>
- De Raat, M., Rood, L., Zuurbier, L., Bloem, B. R., Zuidema, S., & Kunneman, M. (2024). Natural language processing systems for extracting information from electronic health records about activities of daily living: A systematic review. *JAMIA Open*, 7(2), Article ooae044. <https://doi.org/10.1093/jamiaopen/ooae044>
- Eftimov, T., Popovski, G., Petković, M., Seljak, B. K., & Kocev, D. (2024). The Growing Impact of Natural Language Processing in Healthcare and Public Health. *Cureus*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11475376/>
- JMIR Medical Informatics. (2025). Performance of natural language processing for information extraction from electronic health records within cancer: Systematic review. *JMIR Medical Informatics*, 13, Article e68707. <https://doi.org/10.2196/68707>
- King's College Hospital. (2023). Hospital-wide natural language processing summarising the health data of 1 million patients. *PLOS Digital Health*. <https://doi.org/10.1371/journal.pdig.0000218>
- Kondylakis, H., Katehakis, D. G., Kouroubali, A., & Kavlentakis, G. (2023). Natural language processing in electronic health records in relation to healthcare decision-making: A systematic review. *Computers in Biology and Medicine*, 155, Article 106649. <https://doi.org/10.1016/j.compbiomed.2023.106649>
- npj Digital Medicine. (2025). Natural language processing of electronic health records for early detection of cognitive decline: A systematic review. *npj Digital Medicine*, 8, Article 61. <https://doi.org/10.1038/s41746-025-01527-z>

npj Digital Medicine. (2025). Privacy preserving strategies for electronic health records in the era of large language models. *npj Digital Medicine*, 8, Article 16.

<https://doi.org/10.1038/s41746-025-01429-0>

PMC. (2024). Development and validation of natural language processing algorithms in the ENACT national electronic health record research network. *PMC*.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11839006/>

Sänger, M., Grisoni, F., & Schneider, G. (2022). How to keep text private? A systematic review of deep learning methods for privacy-preserving natural language processing. *Artificial Intelligence Review*, 56, 1427–1492. <https://doi.org/10.1007/s10462-022-10204-6>

Zhong, X., Li, S., Chen, Z., Ge, L., Yu, D., Wang, S., You, L., & Shang, H. (2025).

Considerations for patient privacy of large language models in health care: Scoping review. *Journal of Medical Internet Research*, 27, Article e76571.

<https://doi.org/10.2196/76571>